



ME 300: Engineering Thermodynamics

Entropy

Entropy

→ Entropy is a property

→ Classical/Macro: measure of unavailability of thermal energy to do work in a closed system.

→ Statistical/Micro: measure of available molecular energy states at a given T & P

Boltzmann: $S = k \ln \Omega$

classical $\uparrow$ constant. $\uparrow$ energy states

Entropy

$\Rightarrow dS = \frac{\delta Q}{T} \Big|_{\text{reversible}}$

$\delta Q > 0 \rightarrow dS > 0$
 $\delta Q < 0 \rightarrow dS < 0$

$\swarrow$ inexact differential, path dependent
 $\nwarrow$ absolute temp.

$\underbrace{\hspace{1cm}}$
exact
different,
path independent

$$\Rightarrow \Delta S = S_2 - S_1 = \int_1^2 \frac{\delta Q}{T} \Big|_{\text{reversible}}$$

$$\Rightarrow \int_{\text{cycle}} dS = \oint \frac{\delta Q}{T} \Big|_{\text{rev}} = 0 \Rightarrow \oint \frac{\delta Q}{T} \Big|_{\text{rev}} = \frac{Q_H}{T_H} - \frac{Q_C}{T_L} = 0 \quad \frac{Q_H}{Q_C} = \frac{T_H}{T_L}$$

Clausius inequality

↳ whenever a system executes a cycle

$$\oint_{\text{cycle}} \frac{\delta Q}{T} \leq 0 \quad \text{where } = 0 \text{ if qde is reversible}$$

=> 3rd statement

$$\boxed{d\underline{S}_{\text{sys}} + dS_{\text{sur}} \geq 0}$$

Change in entropy

$$\Delta S = \int \frac{\delta Q}{T} \Big|_{\text{rev}} \rightarrow \text{if adiabatic, reversible} \rightarrow \text{isentropic, } \Delta S = 0$$

$\underbrace{\hspace{10em}} \rightarrow S_{\text{generated by irreversibilities}}$

$\Delta S > 0 \rightarrow \text{heat addition (rev, irrev)}$

$\rightarrow \text{adiabatic, irreversible}$

$\Delta S < 0 \rightarrow \text{reversible heat removal}$

$\rightarrow ? \text{ adiabatic? } \times$



The change in entropy for a reversible heat removal process is which of the following?

A. $\Delta S > 0$

B. $\Delta S < 0$

C. $\Delta S = 0$

D. Not enough information

$Q < 0$

Entropy transport

$$\Delta S_{\text{sys}} = \dot{S}_{\text{in}} - \dot{S}_{\text{out}} + \underline{\underline{\dot{S}_{\text{gen}}}}$$

$$\Delta S = \int \frac{\delta Q}{T} + \underbrace{\Phi}$$

entropy generation

due to irreversibility